

Lecture 5 ERS312 Notes

Side Scanning Sonar

- Taking water samples as you go down water columns

Why did Christopher Columbus believe he discovered a sea route to Asia

- Used the wrong circumference
- Wrong information stuck with geographers
- Used it to produce world maps
- Asia was almost at the position of what is North America today

How did Eratosthenes calculate the circumference of the earth?

- Had heard of a well in southern Egypt at the tropic of Cancer
- Sun comes perpendicular
- Positioned a vertical pole (it casted a slight shadow in Alexandria at angle of 7 degrees)
* $360/7=50$

Which ocean current carried the iceberg that the Titanic collided with to the south

- The Labrador current

Satellite and Ship-based Observations

- Could now measure sea surface temperature
- Started in 1970s
- Temperatures have increased very steeply (global warming)
- Temperature recordings by steamships (important because of thermal regulation of the engine)

Sea Surface Temperature Observations

- Most established trade route was in North Atlantic
- Southern Ocean was where no one went by boat (no temperature observations)

Going back in time

- Interested in reconstructing past ocean conditions by analyzing marine sediments
- Hollow tube with heavy lead weight dropped in ocean floor
- Pull it up and you have sediment in the tubing

Sediment Coring

- Youngest layer on top and oldest on the bottom (principle of superposition)
- Retrieve a core and analyze sediment (has all kinds of fossils tells you something about ocean chemistry, temperature, etc.)

Ocean Drilling Program

- A large ship that has a tower with a drill
- Position themselves somewhere on ocean surface
- Put down an array of drill strings
- Average depth of ocean: 3700m
- Penetrates sediment (drills 9m into sediment)
- Project Mohole
 - earth is layered (core, mantle, earth crust)

Project Mohole

- Showed that deep ocean drilling was possible
- In Newfoundland

Mantle Exposure at Gros Morne

- Ocean between North America and Europe (Iapetus ocean)

Wilson Cycle

- The opening and closing of ocean basins
- Takes hundreds of millions of years
- Ocean plate starts going underneath
- Can have mantle rocks squeezed up

